

DAEN 427 Homework 1

Due date: Aug 26, 2025

1. Explain the difference between a *confounder* and a *collider* in the context of a DAG. Why is it important not to condition on a collider when trying to estimate a causal effect?

Answer:

2. You are studying whether a new training program (A) improves job performance (Y). Based on domain knowledge, age and prior experience influence both program participation and job performance. Gender is believed to influence only program participation. Construct a DAG using this information, and identify the minimum set of covariates you would need to adjust for to estimate the causal effect of A on Y .

Answer:

3. Refer to the "Uncomplicated Appendicitis" DAG example in the slides. Discuss the role of *income* and *primary language* in this model. Which one is unmeasured? What potential biases could arise from not including it in your analysis?

Answer:

4. Consider a DOG where a treatment variable A causally affects an outcome Y , and there exists a confounder C that influences both A and Y .
- (a) Explain why the conditional probability $P(Y | A)$ is not equal to the causal effect $P(Y | \text{do}(A))$.
 - (b) Describe how adjusting for C allows us to estimate the causal effect using observational data.

Answer: